DX-SMART

DX-LR22-900T22D
Module technical manual

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深圳大疆无线科技有限公司

Updated records

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V1.0	2025/01/10	Initial version	SML
V1.1	2025/04/19	Optimizing RF parameters	SML
V2.0	2025/06/22	Update the power consumption	SML

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1. Module Introduction

1.1. overview

DX-LR22-900T22D is a low power LoRa module,which is made by SHEN ZHEN DX-SMART TECHNOLOGY CO.,LTD,. For intelligent wireless data transmission, using LLCC68 chip, chip internal integration output maximum 22 DBM power amplifier, anti-jamming performance and further improve communication distance, It has the advantages of low power consumption, high performance and long distance.

1.2. Characteristics of

Performance:

- Output power: + 22 DBM
- Biggest receiving sensitivity: - 134 DBM (BW = 125, SF, = 11)
- Open visual range to 8 km (for reference only and shall be subject to the actual distance to the measured)

Module parameters:

- Operating voltage: 3.3 V to 5.5 V (typical values: 5 V)
- Support the operating frequency range: 850 ~ 949 MHZ
- External antenna
- Operating temperature: -40~+85 ° C

1.3. Basic parameters

Table 1: Table of basic parameters1

The	describe	Notes

parameter name		
Distance of reference	8km	Clear and open environment, antenna height 3 meters, antenna gain 3dbi, level=0
FIFO	230Byte	A single send maximum length
Modulation	LoRa	-
Encapsulation way	Pin type	-
The interface way	Pin insertion	-
Communication port	UART	-
Exterior dimensions	18 (W) (L) x 35.6 x 3.2 mm (H)	Shielding cover included
The radio frequency interface	SMA (outer screw inner hole)	-

1.4. Working parameters

Table 2: Work Parameter Sheet2

Parameter names	details			Notes
	The minimum value	Typical values	Maximum value	
Operating voltage (V)	3.3	5	5.5	5 v or higher can guarantee the output power
Communication level (V)	-	3.3	-	Use 5 v TTL risky burn out
Operating temperature (C)	-40	-	85	Industrial grade design
Operating frequency band (MHz)	850	915	949	Support the ISM
Transmission power	0	-	22	-

(dBm)				
Receiving sensitivity (dBm)	-	-134	-	BW = 125, SF, = 11
Air rate (bps)	336	2148	62500	See the serial port application manual rate level is introduced

Table 3: power consumption table3

Working state	Status	current	Unit
Sleep mode	dormancy	5	uA
Air wake-up mode	Standby	3	mA
	Launch	183	mA
	Receive	9	mA
High time mode	Standby	9	mA
	Launch	184	mA
	Receive	4	mA

Notes

The power consumption parameters are tested under the default parameters of the module.

1.5. Mode of transmission

- Transparent transmission: the sender and the receiver rate level, channel consistent, can transmit data

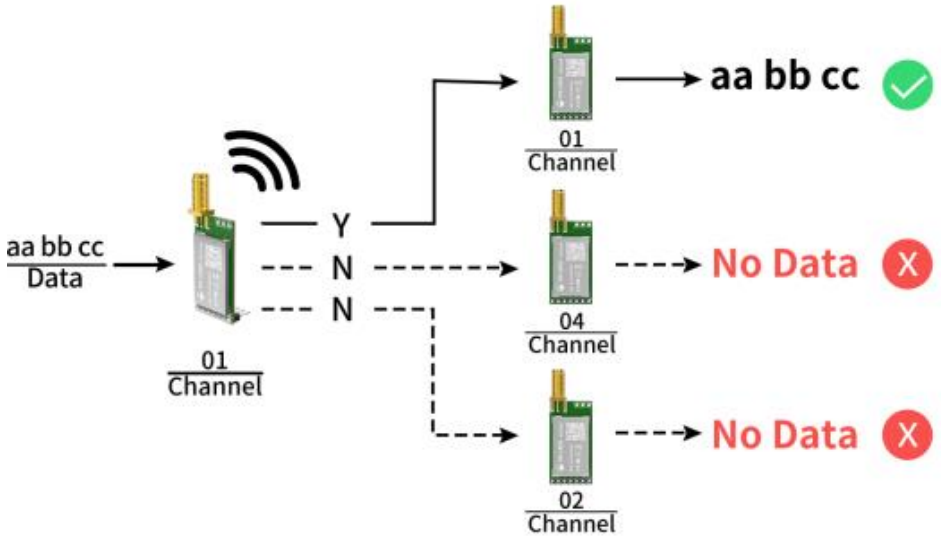


Figure 1: transparent transmission1

- Fixed transfer: the sender to send data, the data contained within the target address, channel to the same as the address at the receiving end, channel, and the sender and the receiver can communication rate level.

Data format, such as: the target address (hexadecimal, two bytes) + target channel (hexadecimal, one byte) + data (hexadecimal)

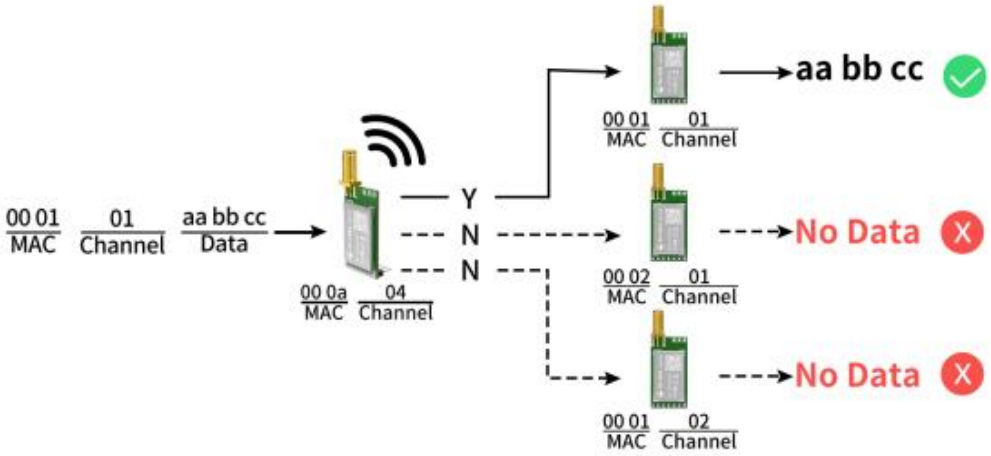


Figure 2: fixed-point transport2

- Broadcast transmission, the sender sends the data, the data within the target channel to the same as the receiver channel, the sender and the receiver rate level and communication.
  Data format, such as: the target channel (one byte, hexadecimal) + data (hexadecimal)

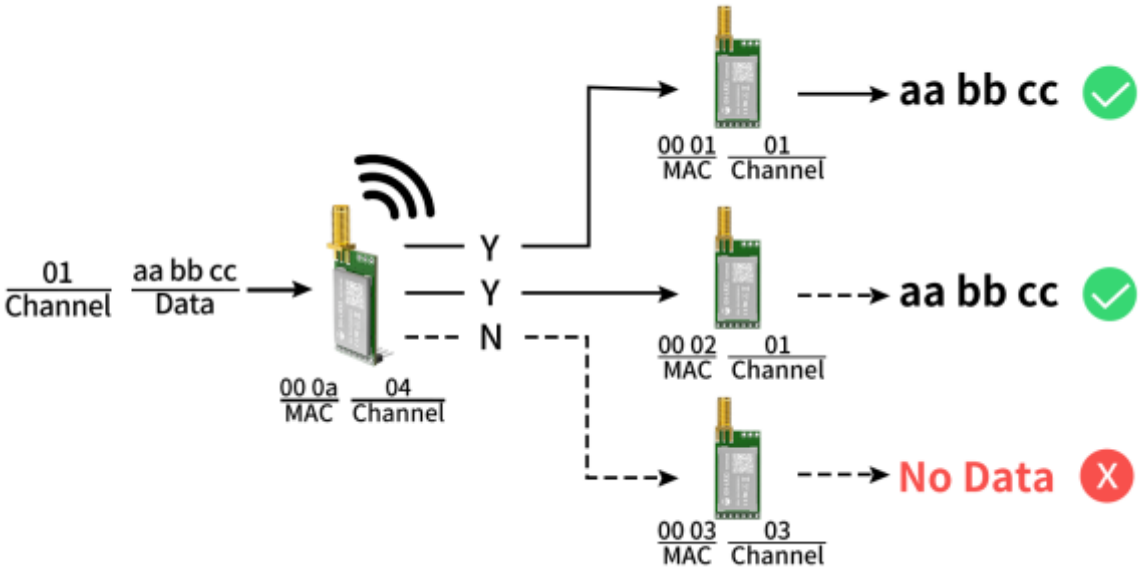


Figure 3: Broadcast transmission

2. Application interface

2.1. Module pin definition

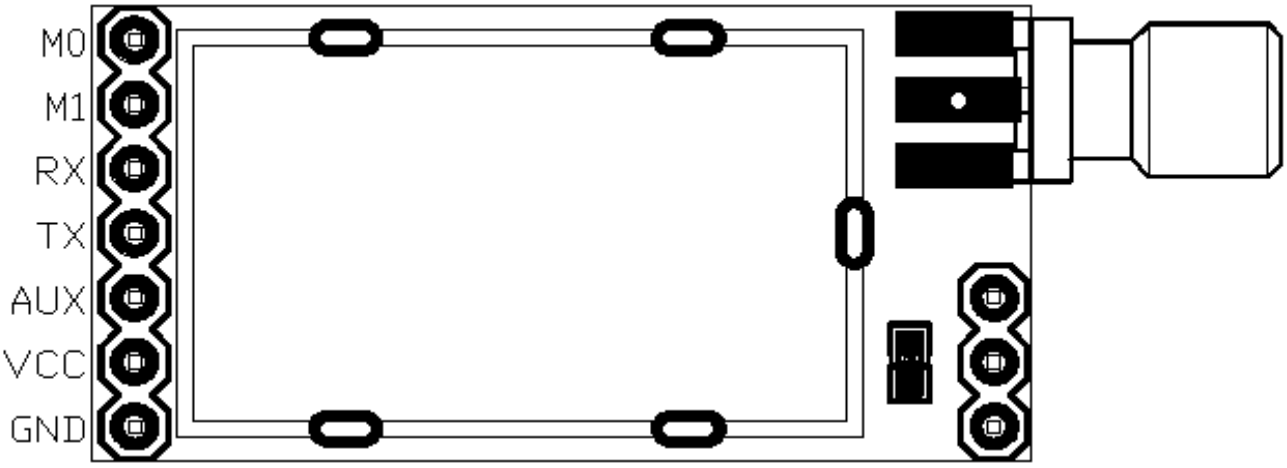


Figure 4: the module's pin definition (front view)4

2.2. Pin definition

Table 4: pin definition table 4

Pin number	Pin name	Pin function	Instructions
1	M0	Input (extremely weak pull-up)	Refer to 2.3.4 for details
2	M1	Input (very weak)	
3	RX	Serial data input	-
4	TX	Serial data output	-
5	AUX	Module RF status indicator foot	Refer to 2.3.3 for details
6	VCC	Power input pin	5V(typical value)
7	GND	Power supply ground	-

2.3. Power supply design

2.3.1. The power interface

Table 5: the power interface pin definition table5

Pin names	Pin number	Description	The minimum value	Typical value	Maximum value	Units
VBAT	6	Power supply module	3.3	5	5.5	V
GND	7	to	-	0	-	V

2.3.2. Power supply stability Requirements

DX - LR22-900 t22d power supply in the range of 3.3 ~ 5.5 V, the need to ensure that not less than 3.3 V input voltage. Below is in the rf burst transmission VVCC_3V3 voltage drop.

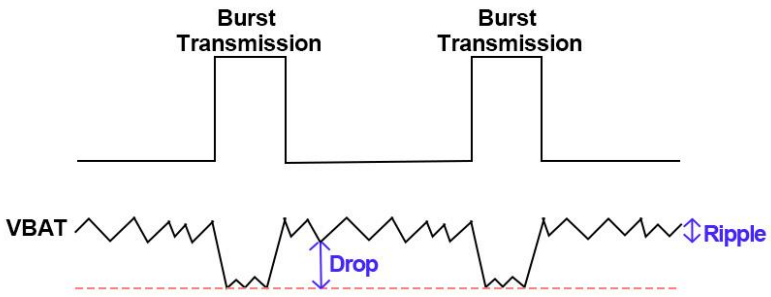


Figure 5: sudden transmission power requirements5

In order to reduce the voltage drop, it is recommended that the reserved for VBAT 2 (100 uF, 0.1 uF) which has the best ESR performance of chip multilayer ceramic capacitors (MLCC), and capacitance placed close to the VBAT pin.

2.3.3. AUX module status indicates the foot work

Pin name	Pin	Description	Pin output	instructions

number			level	
AUX	5	The module status indication	High level	In data sending/data receiving/working mode switching
			Low level	Completion of data/data receiving complete/working mode switching

note

When the detection module to receive data, AUX 2-3 ms, output level in advance for prompt external MCU ready to receive data.

2.3.4. M0, M1 / pin

1. When the command AT + SWITCH = 0, M0 / M1 pin for output, the default output low level, module through the AT command mode SWITCH;
2. When the command AT + SWITCH = 1, M0, M1 / pin for the input, by M0 / M1 pin SWITCH module working mode, working mould

Corresponding to the table below:

M0 input level	M1 input level	Working mode
High level	High level	Sleep mode
High level	Low level	Air wake model
Low level	High level	AT mode
Low level	Low level	High aging model

note

1、Command AT + SWITCH details please refer to the DX - LR22-900 t22d module _ a serial port application guide UART_ 5.3.5 Settings \ query - hardware control pin.
2、When the instruction AT+SWITCH=1, the internal M0/M1 pin is weak pull-up. When switching the working mode of the module through the M0/M1 pin, please be sure to avoid the pin hanging to prevent the abnormal mode switching.

2.4. Refer to the connection circuit

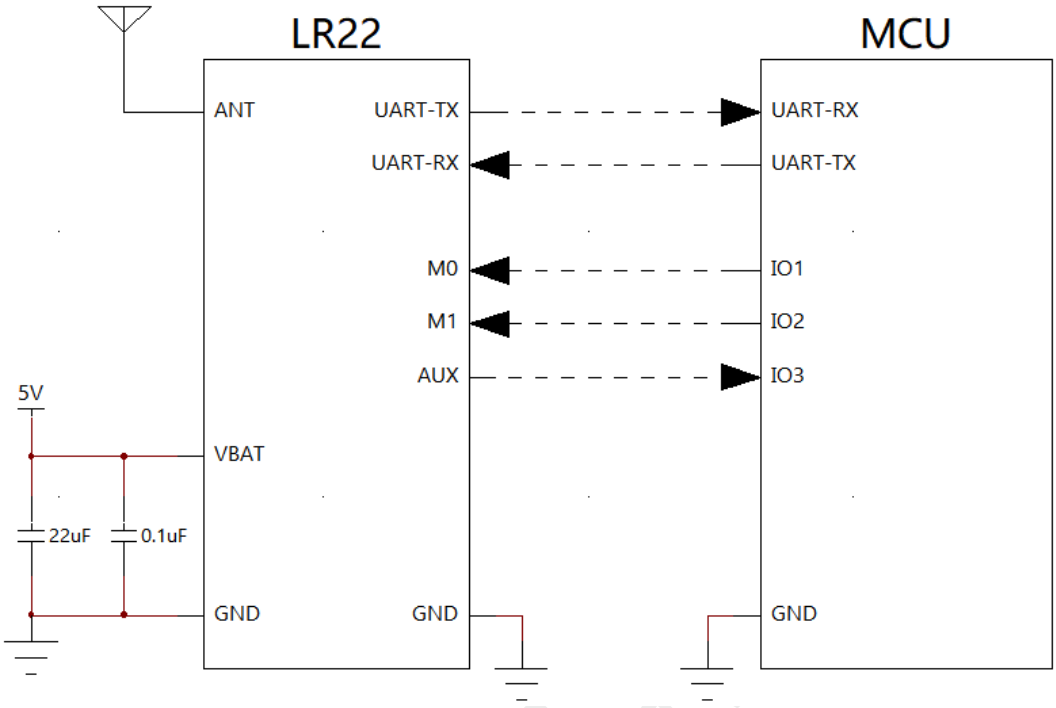


Figure 6: the typical application circuit6

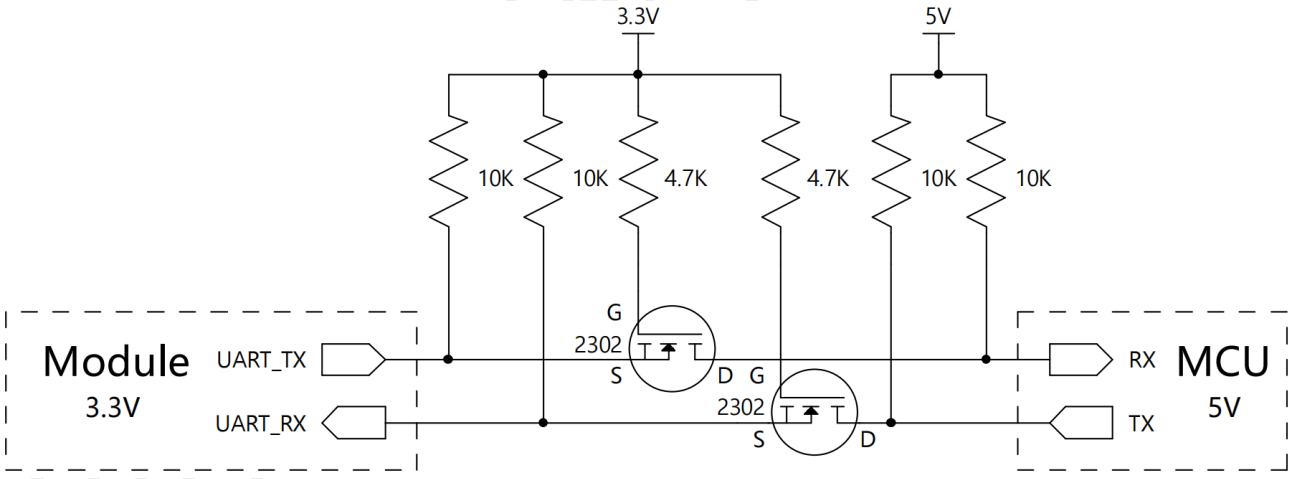


Figure 7: serial level conversion reference circuit7

3. Electrical characteristics, radio frequency characteristics and reliability

3.1. Maximum rating

More than absolute maximum rating pressure may cause permanent damage to the equipment. Absolute maximum rating only on behalf of the tolerance limit, rather than a function. Work need to follow the "recommended operating conditions". Long exposure to the absolute maximum rating conditions may affect the reliability of the equipment.

Table 6: Absolute Maximum Rating Table6

parameters	Minimum	The maximum	Units
VBAT	-0.2	5.5	V
I/O supply voltage (VDDIO)	-0.2	3.7	V
Storage temperature range	-55	+125	℃

Table 7: Recommended conditions of use7

Parameters	Minimum	Typical values	Maximum value	unit
VBAT	3.3	5	5.5	V
I/O supply voltage (VDDIO)	3	3.3	3.7	V
Working temperature range (TA)	-40	+25	+85	℃

3.2. Electrostatic protection

In module applications, such as charged the friction between the human body static electricity, microelectronics produced electrostatic, discharge through various channels to module, may cause certain damage to module, so the ESD protection should be taken seriously. In research, development, production assembly and test process, especially in product design, all ESD protection measures should be taken. For example, in the circuit design of interface and is easily affected by the electrostatic discharge damage or points, should increase the esd protection, anti-static gloves should be worn in production, etc.

Table 8: module for the ESD withstand voltage of pin table8

symbol	describe	The maximum	Units
ESD_HBM	American national standards institute/electronic system design association/electronic equipment manufacturing Chamber of Commerce Standard JS-001-2014 (mannequins) Category 2	+2	kV
ESD_CDM	ESD charging device model, JEDEC standard JESD22-C101D, three levels	+1	kV

4. Mechanical dimensions and layout suggestions

This section describes the mechanical dimensions of modules, all the size of the unit for mm; All did not mark the size of the tolerance, tolerance of + / - 0.3 mm.

4.1. Module machinery '

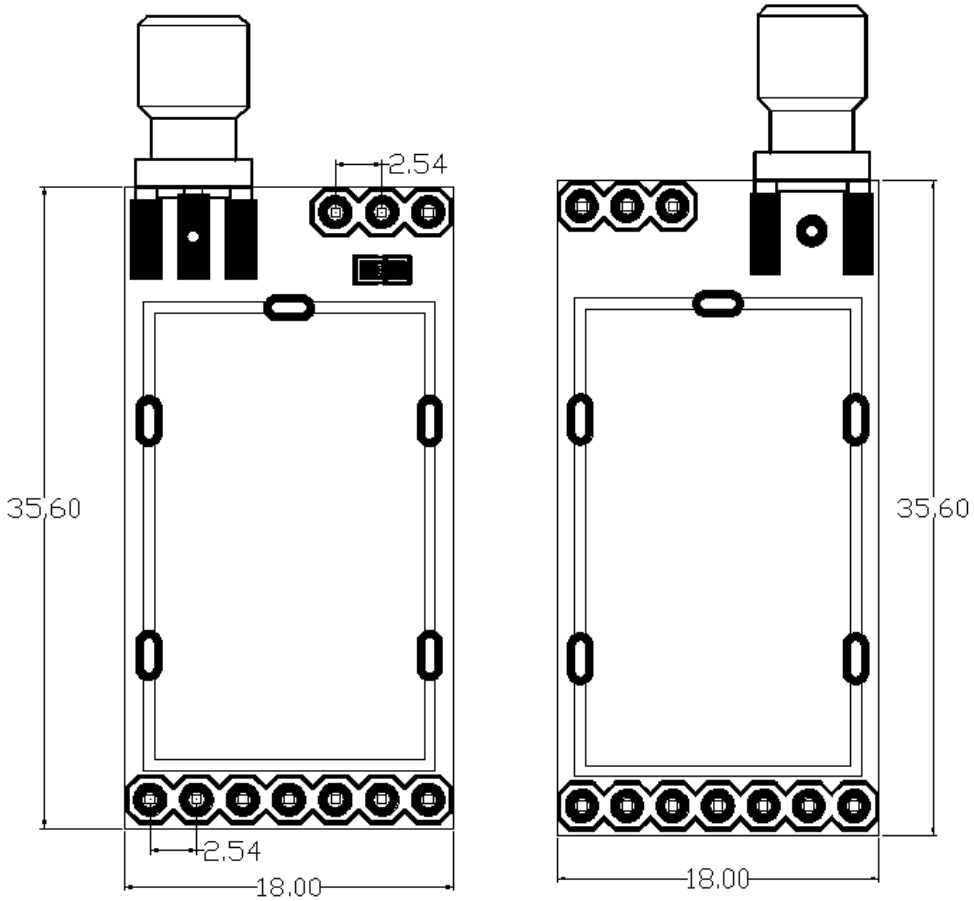


Figure 8: Top view and bottom view dimensions of the module8

4.2. The module top/bottom view

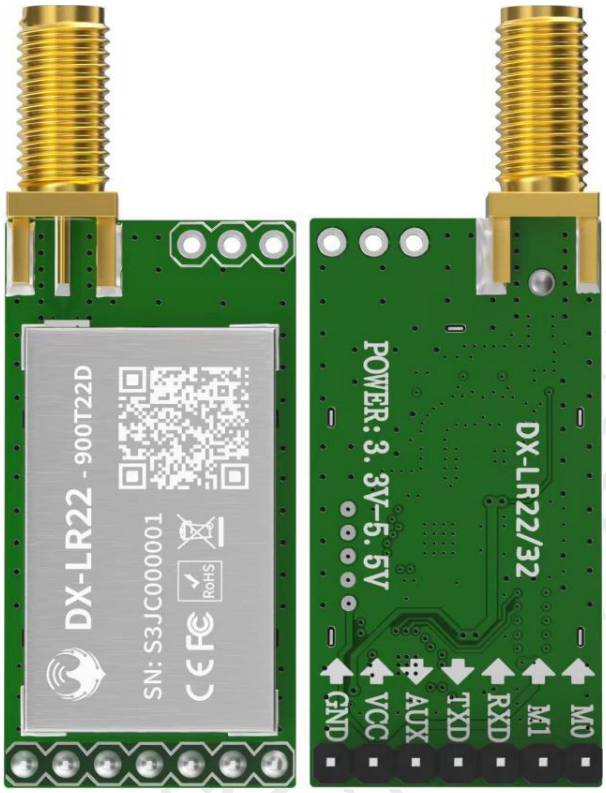


Figure 9: the module's top view and bottom view9

note

Above is for reference only, actual product appearance and label information, please refer to the module object.

4.3. The hardware design layout proposal

DX - LR22-900 t22d modules work in 850-949 - m frequencies, using the external antenna, the antenna standing wave ratio depends on the pattern (VSWR) and efficiency module position, should try to avoid the influence of various factors on the wireless sending and receiving signals, pay attention to the following:

1, surrounded by DX - LR22-900 t22d shell to avoid using metal products, when using part of the metal shell, should try to get module antenna part away from the metal part. Metal wire or metal screws inside the product, should as far as possible away from the antenna module.

2, module antenna should depend on the load board PCB edge place or the load board directly, as far as possible don't placed on board.

3, Suggestions on base board module mounting location use insulating materials in isolation, for example, in the position to put a whole piece of silk screen (TopOverLay).

5. Q&A

5.1. Transmission distance is not ideal

- When there is a straight line communication barriers, communication distance will corresponding attenuation;
- Temperature, humidity, same frequency interference, can lead to communication packet loss rate increase;
- The ground absorption and reflection the radio waves, near ground test effect is poorer;
- Water has a strong ability to absorb the radio waves, so the sea test effect is poor.
- Near the antenna with metal objects, or placed in a metal casing, signal attenuation can be very serious;
- Power register setting error, high air rate setting (the higher the air velocity, the closer distance);
- Power supply voltage is lower than the recommended value at room temperature, the lower the voltage transmission power is smaller;
- Using antenna with poor module matching degree or antenna itself quality problem.

5.2. Module is easy to damage

- Please check the power supply to ensure that between the recommended supply voltage, if more than the maximum value will cause permanent damage to the module;
- Please check the power supply stability and voltage cannot significantly fluctuate frequently;

- Please ensure that the installation process of antistatic operation, the high frequency components electrostatic sensitivity,
- Please ensure that the humidity should not be too high during installation and use, some components are humidity sensitive devices;
- If there is no special demand, it is not recommended to use at too high or too low temperature.

5.3. Bit error rate is too high

- Near the same frequency interference, away from the interference sources or change frequency, channels to avoid interference;
- Power supply is not ideal also may cause garbled, be sure to guarantee the reliability of the power supply;
- Extension line, feeder quality is poor or too long, will also cause a high error rate.

6. Storage and packaging

6.1. Storage conditions

Module in the form of vacuum bags shipment. The module has a humidity sensitivity class of 3 (MSL 3), and its storage is subject to the following conditions:

1. The recommended storage conditions: temperature of 23 + / - 5 ° C, and relative humidity of 35 ~ 60%.

2. Under the recommended storage conditions, module can be in vacuum sealed bags for 12 months.

3. The unsealed workshop life of the module is 168 hours under workshop conditions of 23± 5°C and relative humidity below 60%. Under this condition, can be directly to return or other high temperature operation module. Otherwise, the need to store the module in the relative humidity

is less than 10% of the environment (for example, moisture-proof cabinets) drying in keeping with the module.

4. If the module is in the following conditions, need to pre bake the module in order to prevent the module after hygroscopic be affected with damp be affected with damp and high temperature of welding PCB bubbles, cracks and delamination:

- Storage temperature and humidity is not in conformity with the recommended storage conditions
- Module once opened failed to finish production or storage according to article 3 of the above
- Vacuum packing leakage, material in bulk
- Before the module repair

6.2. Packing Specifications

DX-LR22-900T22D module USES the pallet packing, and vacuum sealing bag packaging, vacuum sealed bags with a desiccant and humidity. Each carrier is 260*150*21.5(unit: mm) and contains 20 modules. The specific specifications are as follows:

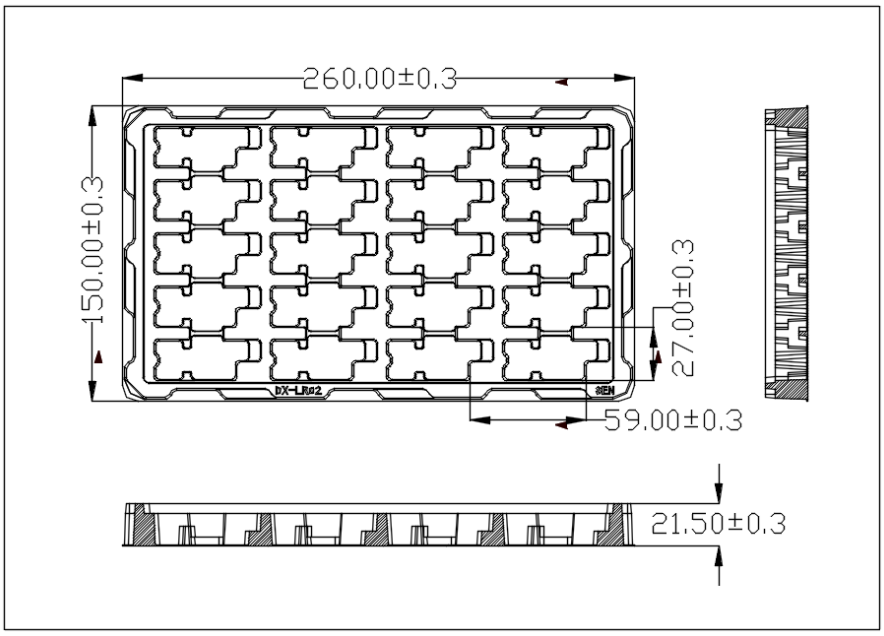


Figure 10: Pallet dimensions (in millimeters)10